



Fig. 8: Structure of perovskite

the capability to produce modules as large rolls of a thin film via high-speed processes.

Solar researchers and industries are confident that the miracle perovskites will soon disrupt the economics of over \$65-billion industry currently dominated by China. Globally, thousands of researchers are involved in the development of perovskite solar cells. The global efforts and the awesome material properties might finally challenge the 60-year reign of crystalline silicon solar cells.

Historically, the lower price tag of emerging photovoltaics has gone hand in hand with significantly lower performance. If we compare power conversion efficiency of third-generation PVs, dye-sensitised solar cells, organic photovoltaics and quantum-dot cells started at just a few per cent and have climbed slowly over the years to roughly 12 per cent. The value for silicon cells is much higher at 20–25 per cent, but perovskite cells may soon give silicon a run for its money with their efficiency soaring to more than 22 per cent in just a couple of years.

Perovskite solar cells use a perovskite structured compound, generally a hybrid organic-inorganic lead or tin halide-based material with lattice-like structure, which helps in light harvesting. Perovskite materials such as methylammonium lead halides are far inexpensive and relatively simple to manufacture. The unique intrinsic properties of perovskites that make them

ideal for solar cell applications are broad absorption spectrum, high absorption coefficient (10^5 cm^{-1}), high dielectric constant, large diffusion length, fast charge separation process, long transport distance of electrons and holes, and long carrier separation lifetime.

As illustrated in Fig. 8, light incident on the transparent electrode of a perovskite solar cell passes onto a light-absorbing perovskite material layer and creates electron-hole pairs (e^-/h^+). The charged particles separate due to low binding energy and eventually diffuse through the charge-conducting layers. The charges are then eventually collected by respective electrodes, producing an electric current.

The World Economic Forum declared perovskite as one of the top 10 emerging technologies of 2016. In the meantime, solar panel manufacturers and frontline universities in the US, Europe and Asia are rushing to commercialise this technology.

The perovskite mineral is named after a Russian mineral expert Lev Perovski, who first studied it. Later, researchers found that mineral deposits containing perovskite structures were cheap and abundant throughout the world. Its structure remained poorly understood and ignored for many years as the industry was already biased that silicon was the best material for solar cell applications.

Scientists weren't sure about perovskite's usefulness until 2009,

when a Japanese researcher Tsutomu Miyasaka, a professor at Tohoku University of Yokohama, found that perovskite could absorb sunlight and turn it into electricity.

Perovskite presents significant opportunities to realise a low-cost, industry-scalable solar cell technology. This potential for low cost and scalability requires overcoming barriers related to stability and environmental compatibility. If these issues are overcome, a perovskite-based technology could be utilised for terawatt-scale solar deployment.

Scientists are now experimenting with a hybrid structure made of organic molecules and inorganic elements within a single crystalline structure, which together capture light and convert it into electricity. Researchers have achieved 26 per cent efficiency by mechanically combining perovskite with silicon solar cells. These devices use compounds with perovskite crystal structure and stoichiometry to absorb light. $(\text{CH}_3\text{NH}_3)\text{PbI}_3$ is the most studied example. Scientists do not have clear understanding about their operation mechanism and efficient and fast charge transfer mechanism. Yet, the remarkable rate of progress caused photovoltaics researchers in academia and the industry to switch to perovskites.

Chemicals and materials for perovskite solar-cell R&D are sold by many companies, but none has perovskite modules for sale. Saule Technologies has demonstrated